

Fundamentals of Cyber Defense & SOC Operations

Phase 2 (Log Management & SIEM Essentials)

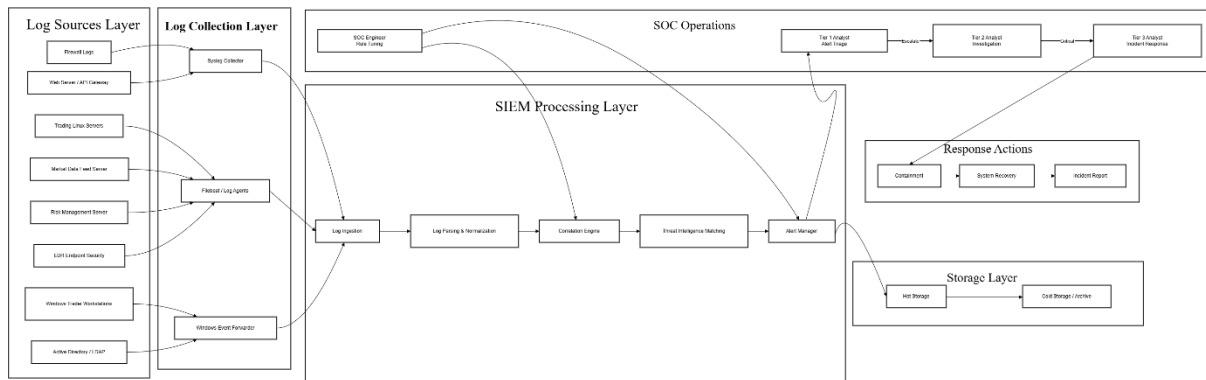
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1. Log Source → SIEM Mapping Table

Log Source	Device/System	Log Type	Security Use Case	Why Needed in SIEM
Firewall	Perimeter Firewall	Network traffic logs	Detect DoS and unauthorized connections	Detect external attacks
Trading Servers	Linux Servers	Syslogs	Detect abnormal trading activity	Identify algorithm compromise
Trader Workstations	Windows OS	Security logs	Detect insider threats	Monitor login misuse
Order Execution Engine	Trading Application	Application logs	Detect order manipulation	Maintain trade integrity
Risk Management System	Risk Server	Audit logs	Detect circuit breaker triggers	Prevent flash crash
Market Data Feed Server	Data Server	Application logs	Detect data tampering	Ensure data integrity
Active Directory	Authentication Server	Authentication logs	Detect brute force attacks	Monitor access control
Endpoint Security	EDR	Malware alerts	Detect malware infection	Endpoint protection
Web Gateway	Proxy/Web Server	HTTP logs	Detect API abuse	Detect unauthorized trading API usage
SIEM Platform	SOC System	Correlated alerts	Incident investigation	Central monitoring

2. SIEM Architecture (Conceptual Design)



SIEM Architecture Explanation

Role of Log Collectors

Log collectors act as intermediaries between log sources and the SIEM platform. Their main role is to gather logs from different systems such as trading servers, trader workstations, firewalls, market data servers, and authentication systems, and forward them securely to the SIEM for analysis.

In the HFT trading environment, multiple critical systems generate logs continuously. For example, trading servers generate Linux syslogs, trader systems generate Windows security logs, and firewalls generate network traffic logs. Without collectors, managing logs from these heterogeneous sources would be difficult.

Collectors also perform:

- Log aggregation from multiple devices
- Secure log forwarding
- Filtering unnecessary logs
- Reducing network overhead
- Ensuring reliable log delivery

In this project, tools like syslog collectors, Windows event forwarders, and log agents (like Filebeat) ensure that all infrastructure logs reach the SIEM in near real-time so that suspicious trading activity or cyber attacks can be detected quickly.

Why Log Normalization is Required

Different devices generate logs in different formats and structures. For example, a failed login event in Windows appears as an Event ID, while in Linux it appears as a syslog message. Firewall logs also use different formats.

Because of this inconsistency, direct comparison of logs becomes difficult. Log normalization solves this problem by converting logs into a standard format so they can be analyzed together.

For example:

Windows log:

EventID 4625 – Failed Login

Linux log:
Failed password for invalid user

After normalization:

Event Type: Authentication Failure
Severity: Medium
Source: Host System

Normalization helps SIEM to:

- Standardize log formats
- Enable easier searching
- Improve correlation accuracy
- Reduce analysis complexity
- Improve dashboard visualization

In the HFT trading SOC, normalization helps correlate events such as login failures, algorithm changes, and unusual trading activity across different platforms.

What the Correlation Engine Does

The correlation engine is the intelligence component of the SIEM. It analyzes normalized logs and identifies patterns that may indicate security incidents.

Instead of looking at individual events, the correlation engine connects multiple related events to detect sophisticated attacks.

For example:

Single event:

One failed login attempt → normal behavior

Correlated events:

Multiple failed logins + privileged access + algorithm modification

Result:

Possible insider manipulation detected.

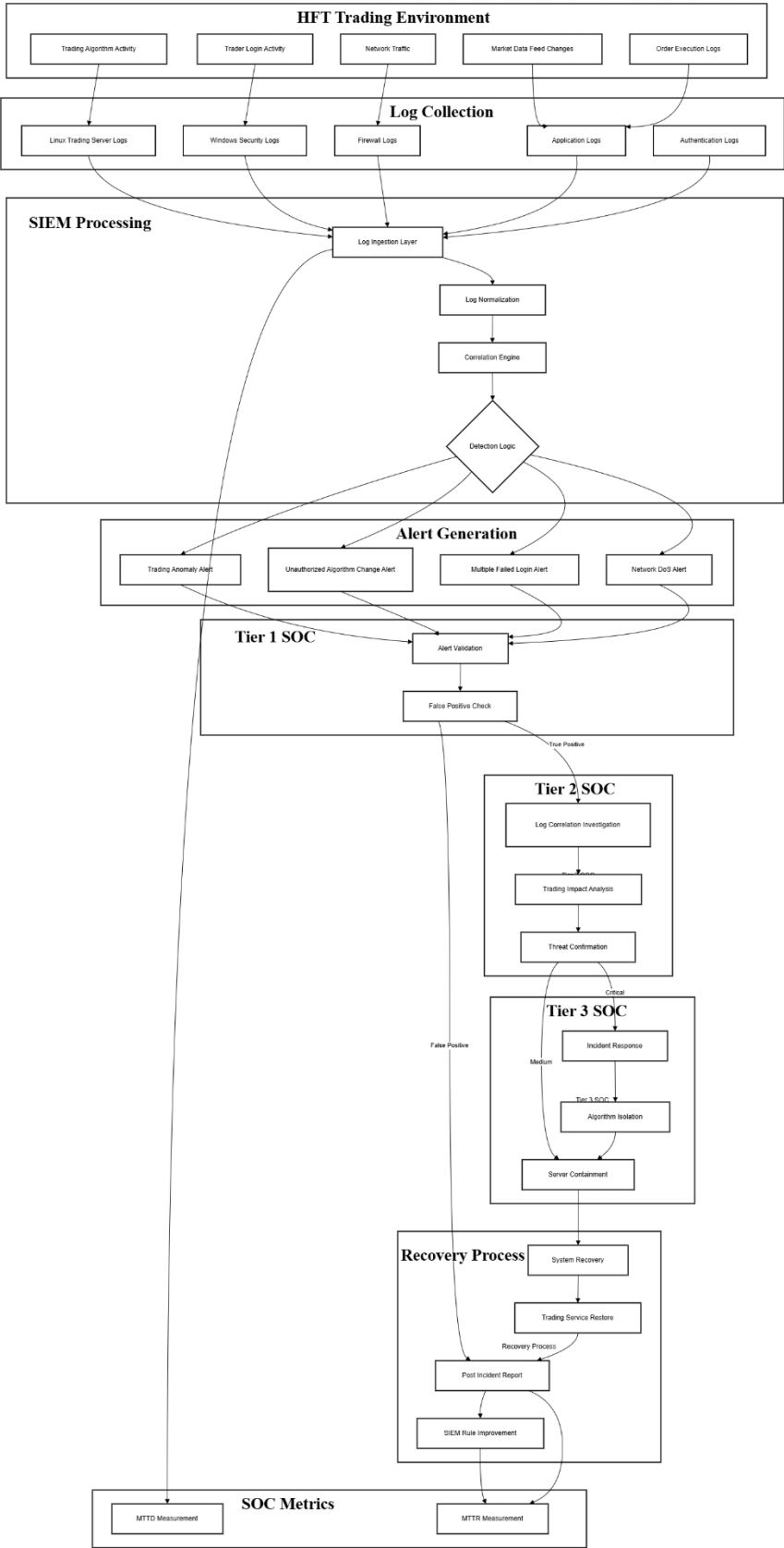
The correlation engine works using predefined rules and behavioral patterns. In this project, correlation helps detect:

- Unauthorized algorithm modification
- Market data manipulation
- Distributed denial-of-service attacks
- Insider trading misuse
- Brute force login attempts

When suspicious patterns are detected, the SIEM generates alerts which are then investigated by SOC analysts.

Thus, the correlation engine enables proactive threat detection instead of reactive monitoring, which is critical in high-frequency trading environments where even milliseconds of delay can cause financial loss.

3. Alert Lifecycle Flow



Alert Lifecycle Explanation in HFT Trading SOC

In the High-Frequency Trading (HFT) SOC environment, alerts follow a structured lifecycle to ensure rapid detection and response to security incidents that may impact trading integrity and market stability.

Stage 1: Log Generation and Collection

The alert lifecycle begins when logs are generated from various components such as trading algorithms, market data servers, trader workstations, firewalls, and authentication systems. These logs are collected and forwarded to the SIEM platform through log collectors.

Responsible team:
IT infrastructure and automated SIEM ingestion systems.

The **Mean Time to Detect (MTTD)** measurement begins at this stage, as it represents the time taken from log generation to alert detection.

Stage 2: SIEM Detection and Alert Generation

The SIEM platform performs log ingestion, normalization, and correlation. If suspicious patterns are detected, such as multiple failed trader logins, unusual trading activity, or unauthorized algorithm modifications, the SIEM generates alerts.

Responsible system:
SIEM detection engine.

At this stage, MTTD continues to be measured until the alert is generated.

Stage 3: Tier-1 SOC Alert Triage

The generated alert is first handled by the Tier-1 SOC analyst. Their responsibility is to validate the alert, check basic indicators, and determine whether it is a false positive or a genuine security incident.

Responsible team:
Tier-1 SOC analysts.

False positives are identified and closed at this stage after documentation. This helps reduce alert fatigue and allows the SOC team to focus on real threats.

Stage 4: Tier-2 SOC Investigation

If the alert is confirmed as a true positive, it is escalated to the Tier-2 SOC analysts. They perform detailed investigation by correlating logs, analyzing trading impact, and validating whether the activity indicates market manipulation, insider abuse, or infrastructure attacks.

Responsible team:
Tier-2 SOC analysts.

If the issue is moderate severity, Tier-2 may initiate containment actions such as blocking IP addresses or disabling suspicious user accounts.

MTTR measurement typically begins at this stage because incident response activities start here.

Stage 5: Tier-3 Incident Response

Critical incidents are escalated to Tier-3 SOC analysts. They handle advanced incident response activities such as forensic analysis, algorithm isolation, server containment, and coordination with the trading operations team.

Responsible team:
Tier-3 SOC analysts and SOC engineers.

This stage focuses on minimizing financial risk and restoring secure trading operations.

Stage 6: Recovery and Closure

After containment, systems are recovered, trading services are restored, and incident documentation is completed. SIEM rules may also be updated to improve future detection capability.

Responsible team:
Tier-3 SOC, SOC engineers, and operations team.

The **Mean Time to Respond (MTTR)** is measured from the start of investigation until the incident is fully resolved and closed.

4. Three Sample Correlation Rules

Rule 1 – Authentication Based Correlation Rule

Rule Name:
Unauthorized Trader Account Access Detection

Log Sources Used:
Windows Security Logs (Trader Workstations)
Active Directory Authentication Logs
SIEM Authentication Alerts

Condition / Logic:

If more than **5 failed login attempts** occur for a trader account within **10 minutes**

AND

Login attempts originate from an **unrecognized workstation or IP address**

AND

A successful login occurs immediately after failed attempts

THEN

Generate a **HIGH severity alert** and notify SOC Tier-1.

Why This Rule Is Important:

This rule helps detect unauthorized access attempts targeting trader accounts which could allow attackers or insiders to manipulate trading systems.

Threats Detected:

- Brute force attack on trader accounts
 - Compromised trading credentials
 - Insider misuse of accounts
 - Unauthorized access to trading systems
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Rule 2 – Network Based Correlation Rule

Rule Name:

Trading Infrastructure Availability Attack Detection

Log Sources Used:

Firewall Logs
Network Traffic Logs
Trading Server Performance Logs
SIEM Network Alerts

Condition / Logic:

If inbound network traffic to trading servers exceeds the normal baseline threshold **AND**

More than **100 connection attempts** occur from the same IP within **5 minutes**

AND

Trading order execution latency increases beyond acceptable threshold

THEN

Generate a **CRITICAL severity alert** and escalate to Tier-2 SOC.

Why This Rule Is Important:

In an HFT environment, even small delays can cause major financial loss. This rule helps detect attempts to disrupt trading availability through network flooding or targeted infrastructure attacks.

Threats Detected:

- Denial-of-Service (DoS) attacks
- Trading infrastructure disruption attempts

- Network congestion attacks
 - Market timing manipulation attempts
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Rule 3 – Endpoint / Trading System Integrity Rule

Rule Name:

Trading Algorithm Integrity Violation Detection

Log Sources Used:

Linux Trading Server Syslogs
Algorithm Application Logs
Endpoint Detection & Response (EDR) Alerts
File Integrity Monitoring Logs

Condition / Logic:

If a trading algorithm file is modified

AND

The change is not performed by an authorized developer account

AND

Modification occurs during active trading hours

AND

EDR detects suspicious process execution

THEN

Generate a **CRITICAL severity alert** and escalate to Tier-3 SOC.

Why This Rule Is Important:

Trading algorithms are critical assets in HFT firms. Unauthorized modification may result in flash crashes, market manipulation, or financial loss.

Threats Detected:

- Algorithm manipulation attacks
- Insider trading abuse
- Malware tampering of trading software
- Unauthorized code modification
- Flash crash risk due to algorithm changes